UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

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Abstract

Vision-language-action models inherit strong semantic priors from large vision-language pre-training, but they often remain weak at the low-level spatial reasoning and short-horizon physical dynamics needed for precise robot manipulation. We present UP-VLA, a unified autoregressive policy that jointly learns multimodal understanding, future visual prediction, and action generation within one backbone. UP-VLA instantiates this idea with a Phi-1.5 language model backbone, a continuous CLIP vision stream for multimodal understanding, and a discrete MAGVIT token stream for future-image prediction. A task-aware prompting scheme and attention masks allow the same transformer to handle image-grounded question answering, future visual token prediction, and multi-step action decoding. Training combines LLaVA-style multimodal understanding data with robot demonstration data, then fine-tunes the policy while retaining the understanding objective. On CALVIN, UP-VLA reaches an average completion length of 4.08 on ABC→D and 4.42 on ABCD→D, outperforming strong VLA and prediction-based baselines including Robo-Flamingo and GR-1. In real-world Panda experiments, the full model achieves 0.80 success on seen tasks and 0.58 on unseen-object tasks. Ablations show that both predictive pre-training and multimodal understanding are important: removing prediction reduces ABC→D performance from 4.08 to 1.44, while removing multimodal understanding sharply degrades unseen-object success from 0.58 to 0.20.

1. Introduction

Large pre-trained vision-language models have changed the design of embodied policies. By transferring semantic knowledge from internet-scale image-text corpora, recent vision-language-action (VLA) systems improve instruction following and open-vocabulary generalization (Driess et al., 2023; Li et al., 2023b; Zhen et al., 2024). However, strong semantics alone do not guarantee accurate control. Manipulation tasks also depend on geometry, contact-sensitive state changes, and short-horizon dynamics, all of which are only weakly supervised by conventional visual question answering or captioning objectives. This mismatch is particularly costly in long-horizon benchmarks such as CALVIN (Mees et al., 2022), where small spatial errors early in a sequence compound across multiple subtasks.

Prediction-based robot policies offer a complementary inductive bias. By learning to forecast future observations or denoise future actions, they can encode physical regularities that help planning and control (Wu et al., 2023; Black et al., 2023; Chi et al., 2023; Ke et al., 2024). Yet these models often lag behind VLM-based policies in semantic grounding, or they require separate modules for language understanding, subgoal generation, and action execution. This separation weakens information sharing and increases system complexity.

This paper studies whether these two strengths can be combined within one model. Our central hypothesis is that a single autoregressive policy can become both semantically grounded and dynamics-aware if it is trained jointly on multimodal understanding and future visual prediction, and if these tasks are expressed through a unified prompting interface. Based on this idea, we introduce UP-VLA, a unified understanding-and-prediction VLA model built on a Phi-1.5 backbone (Li et al., 2023c). UP-VLA uses a CLIP vision tower (Radford et al., 2021) for multimodal understanding and a MAGVIT tokenizer (Yu et al., 2023) for discrete future-image prediction. A task token and task-specific attention mask determine whether the model solves an understanding problem or predicts future visual tokens and actions from current observations.

Our implementation therefore co-trains multimodal understanding on LLaVA-style data (Liu et al., 2024) and future-view prediction on robot demonstrations, then fine-tunes the policy with joint future-image and action supervision while retaining the understanding objective.

Our contributions are threefold. First, we propose a unified training formulation that jointly optimizes multimodal understanding, future prediction, and action learning in one autoregressive backbone. Second, we design task-aware prompting and attention that switch between understanding and prediction without separate expert modules. Third,

we show strong CALVIN gains and improved real-world unseen-object manipulation, with ablations isolating the roles of multimodal understanding, future prediction, and Bridge pre-training.

2. Related Work

2.1. Vision-language-action models

Recent embodied foundation models adapt large vision-language models to robot control by conditioning action prediction on language and visual context (Driess et al., 2023; Brohan et al., 2022; Li et al., 2023b; Zhen et al., 2024). PaLM-E (Driess et al., 2023) and RT-1 (Brohan et al., 2022) show that scaling multimodal pre-training and robot data improves generalization, while Robo-Flamingo (Li et al., 2023b) demonstrates that lightweight imitation learning on top of a pre-trained VLM can already be competitive on manipulation benchmarks. These methods, however, primarily optimize action prediction or instruction following. They do not explicitly force the policy to model short-horizon future observations, which can matter for precise manipulation. UP-VLA targets this gap by co-optimizing semantic understanding and predictive modeling in the same backbone.

2.2. Predictive policies and world models

Prediction has become a central ingredient in robot learning. GR-1 (Wu et al., 2023) jointly predicts future images and actions with a GPT-style policy, SuSIE (Black et al., 2023) uses predicted visual subgoals from image-editing diffusion, and diffusion-based action models such as Diffusion Policy (Chi et al., 2023) and 3D Diffuser Actor (Ke et al., 2024) show strong robustness and multimodal action modeling. World-model-style embodied systems such as 3D-VLA (Zhen et al., 2024) further argue that action generation should be mediated by predictive representations. UP-VLA is closest in spirit to GR-1 (Wu et al., 2023), but differs in two important ways: it explicitly preserves multimodal understanding through joint co-training on LLaVA-style data, and it uses a unified prompting interface derived from multimodal understanding/generation models rather than a control-only predictive setup.

2.3. Unified multimodal modeling

General multimodal models increasingly blur the line between understanding and generation. BLIP-2 (Li et al., 2023a), InstructBLIP (Dai et al., 2024), and LLaVA (Liu et al., 2024) focus on strong multimodal understanding, while Show-o (Xie et al., 2024) demonstrates that a single transformer can support both multimodal understanding and generation. UP-VLA borrows this unification perspective for robotics. Unlike these general-domain models, however, our generation target is not free-form image synthesis but embodied future-state prediction aligned with robot actions. This distinction matters because robot control requires predictions that remain tightly coupled to executable motion rather than visually plausible but behaviorally irrelevant generations.

Figure 3 highlights the difference between VLM-based VLAs, predictive policies, and the unified UP-VLA design.

3. Method

3.1. Problem setup

We study language-conditioned visuomotor control from current observations. At each step, the policy receives an instruction and one or more RGB observations of the scene, and it must predict both a future visual state and a sequence of low-level actions. The released training configuration uses a horizon of ten future action steps and predicts future visual tokens for one camera view. Evaluation focuses on long-horizon zero-shot task chaining in CALVIN (Mees et al., 2022) and a real Panda manipulation setup.

3.2. Model architecture

UP-VLA is built on a Phi-1.5 decoder-only language model (Li et al., 2023c). The model has two visual pathways with different roles:

1. A continuous multimodal understanding (MMU) stream uses a frozen CLIP ViT-L/14-336 encoder (Radford et al., 2021) followed by a learned projector into the language-model embedding space. This stream is used for image-grounded question answering and scene understanding.
2. A predictive (PRE) stream uses a frozen MAGVIT tokenizer (Yu et al., 2023) to convert robot observations and future observations into discrete visual tokens. These tokens are consumed and predicted autoregressively by the shared transformer.

For action prediction, the model appends ten special `<|lvg|>` tokens after the predictive sequence. Their final hidden states are pooled by a lightweight token learner and mapped to ten 7-dimensional actions by an MLP head. In the released implementation, the action head is trained with mean-squared error against relative actions over the next ten control steps.

3.3. Unified prompting and attention

The key mechanism in UP-VLA is a task-aware prompting scheme that uses the same transformer for understanding and prediction. Figure 2 illustrates the overall pipeline, and Figure 4 shows the prompting pattern.

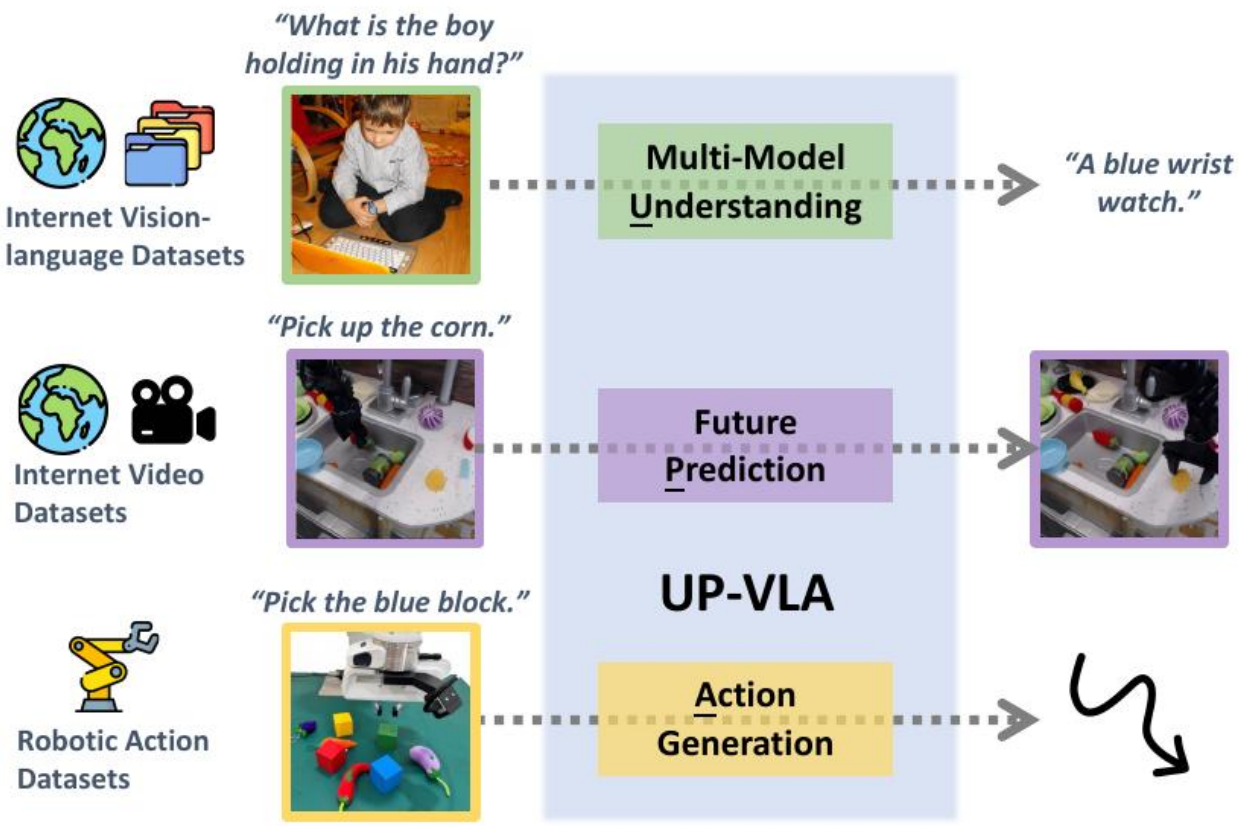


Figure 1. Motivation for UP-VLA. Understanding, future prediction, and action generation are trained in one model.

For predictive training, the prompt begins with <|t2i|>, followed by the instruction text, then <|soi|> and the current image tokens, and ends with <|eoi|> plus ten <|lvg|> placeholders. The model is trained to predict future image tokens in the visual span and future actions from the <|lvg|> states. The attention mask keeps text causal, while allowing the image segment to be jointly visible, matching the implementation in the released prompting utilities. Intuitively, this lets the model reason over the full current observation while still decoding future content autoregressively.

For multimodal understanding, the prompt begins with a dedicated <|mmu|> token, places projected CLIP image embeddings between <|soi|> and <|eoi|>, and appends the text question and answer tokens. The corresponding mask exposes the image embeddings to the text decoder while preserving autoregressive answer generation. This yields a standard image-conditioned generation mode without requiring a separate VQA model.

3.4. Training objectives

UP-VLA is trained in two stages.

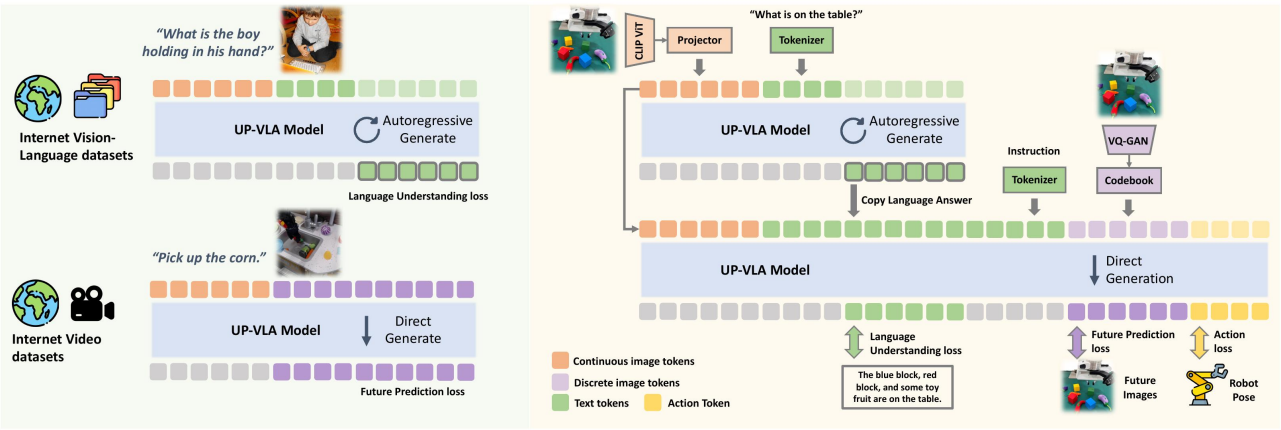
Stage 1: prediction-understanding pre-training. We jointly optimize future-view prediction on robot demonstrations and multimodal understanding on the LLaVA-665k instruction-tuning corpus (Liu et al., 2024). The released configuration also supports BridgeData (Walke et al., 2023) pre-training for the predictive stream, and the ablation study confirms that removing Bridge pre-training causes a large downstream drop.

Stage 2: action tuning with continued MMU co-training. We fine-tune on downstream robot data while keeping the MMU objective active. The predictive branch now supervises both future visual tokens and the ten-step action sequence.

Let $\mathcal{L}_{\text{pre}}$ denote cross-entropy over future visual tokens, $\mathcal{L}_{\text{mmu}}$ denote cross-entropy over answer text in MMU mode, and $\mathcal{L}_{\text{act}}$ denote mean-squared error on the action sequence. The training loss is

$$\mathcal{L} = \lambda_{\text{pre}}\mathcal{L}_{\text{pre}} + \lambda_{\text{act}}\mathcal{L}_{\text{act}} + \lambda_{\text{mmu}}\mathcal{L}_{\text{mmu}}.$$

In the released configuration, $\lambda_{\text{pre}} = 1.0$, $\lambda_{\text{mmu}} = 0.1$, and λ_{act} is set to 0 during prediction pre-training and 1.0 during action tuning.



(a) Pretraining Stage (b) Finetuning Stage

Figure 2. Overview of UP-VLA. A multimodal understanding stream and a future-prediction-and-action stream share one autoregressive transformer. Current observations and language are encoded into a shared token space, and future visual and action tokens are decoded with task-aware prompting.

3.5. Implementation details

The default configuration uses 1024 visual tokens per image, image resolution 512×512, a maximum text length of 576 tokens, and a ten-step action horizon. MMU uses CLIP image preprocessing at 336×336, while the predictive branch operates on discrete MAGVIT tokens. Training is implemented with mixed precision and gradient accumulation. During rollout, the policy predicts ten actions at a time, executes them open loop, then replans from the next observation.

4. Experiments

4.1. Experimental setup

Datasets. For multimodal understanding, we use the released LLaVA instruction-tuning data (Liu et al., 2024). For robot prediction and control, we use CALVIN (Mees et al., 2022); the implementation also supports BridgeData-style pre-training (Walke et al., 2023). Real-world evaluation is conducted on a Franka Panda platform with seen and unseen-object tasks.

Benchmarks. We report long-horizon zero-shot evaluation on two CALVIN settings. ABC→D evaluates transfer to unseen environments after training on subsets A, B, and C. ABCD→D evaluates zero-shot chaining within the full CALVIN training split and tests generalization to environment D. Following prior work, performance is summarized by success rates for completing 1 to 5 instructions in sequence and by average successful sequence length.

Baselines. We compare against RT-1 (Brohan et al., 2022), Diffusion Policy (Chi et al., 2023), Robo-Flamingo (Li et al., 2023b), GR-1 (Wu et al., 2023), SuSIE (Black et al., 2023), 3D-VLA (Zhen et al., 2024), and 3D Diffuser Actor (Ke et al., 2024) where reported in the provided evaluation tables.

4.2. Main results on CALVIN

UP-VLA achieves the strongest results on the harder ABC→D transfer setting, reaching an average sequence length of 4.08 and a 5-task success rate of 0.699. This is a clear improvement over GR-1's 3.06 average length and 0.401 5-task success, and it substantially exceeds the VLM-based Robo-Flamingo baseline, which reaches 2.47 average length. The gain over the reproduced UP-VLA-RT-2 variant without prediction is especially large, suggesting that semantic grounding alone is insufficient for robust long-horizon control.

The same pattern holds on ABCD→D. UP-VLA attains the best performance at every chain length from 2 through 5, with an average completion length of 4.42 and 5-task success of 0.812. The margin over GR-1 is smaller than in ABC→D, but still consistent.

Comparing the two CALVIN settings clarifies where the method helps most. The gain over GR-1 is much larger on the harder ABC→D transfer split (+1.02 average length and +0.298 at 5 tasks) than on ABCD→D (+0.21 average length and +0.081 at 5 tasks). This is consistent with the role of predictive supervision: when environment shift is stronger, modeling future visual consequences becomes more valuable.

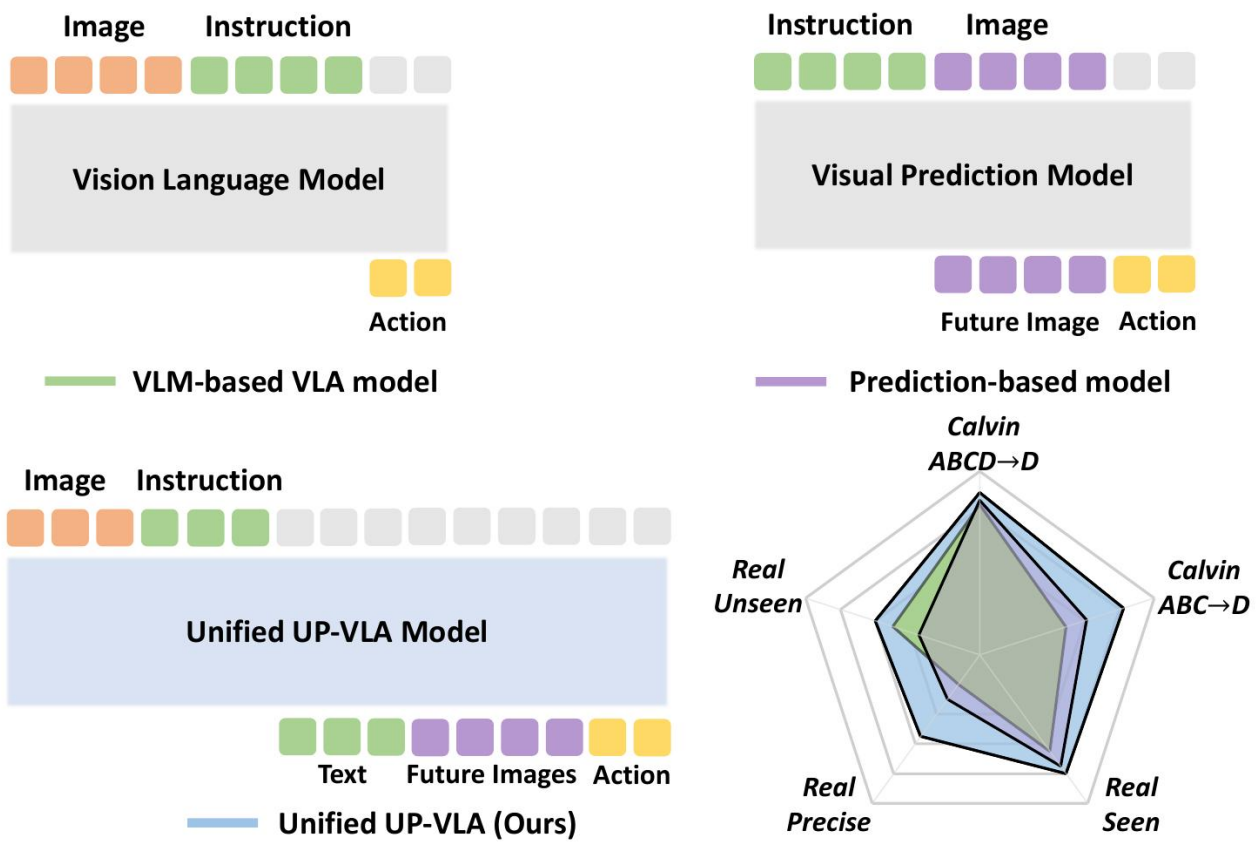


Figure 3. Positioning of UP-VLA relative to VLM-based and prediction-based robot policies. UP-VLA keeps text understanding, future-image prediction, and action decoding in one backbone.

4.3. Real-world evaluation

The real-world results are more modest than the CALVIN gains, but they support the same conclusion. The full model achieves 0.80 success on seen tasks and 0.58 on unseen-object tasks, indicating that retained multimodal understanding helps generalization under appearance shift.

The split between seen and unseen objects is diagnostically useful. Seen-task success remains competitive even for some weakened variants, whereas unseen-object success drops sharply when MMU is removed. This implies that MMU is doing more than generic regularization: it helps bind language to object attributes that are weakly covered by robot demonstrations, which becomes important once the evaluation distribution shifts in appearance.

5. Analysis

5.1. Ablation study

The ablations in Table 3 show that the three ingredients of UP-VLA are complementary.

Removing future prediction causes the largest degradation on CALVIN, dropping ABC→D average length from 4.08 to 1.44. This supports the claim that predictive supervision is not just an auxiliary loss but a core source of control-relevant spatial and dynamical information.

Removing MMU has a smaller effect on CALVIN average length, which falls to 3.89, but it sharply reduces unseen-object real-world success from 0.58 to 0.20. This asymmetry is important: semantic understanding contributes most when the model must transfer beyond the objects and appearances seen in robot demonstrations.

Removing Bridge pre-training also produces a large drop, down to 2.74 average length on ABC→D and 0.30 unseen-object success. This indicates that predictive pre-training on broader robot interaction data remains useful even when the final policy is tuned on CALVIN. Finally, removing MMU conditioning leaves CALVIN performance almost unchanged but lowers unseen-object success from 0.58 to 0.50, suggesting that explicit language-side scene information mainly helps under appearance shift rather than in-distribution control.

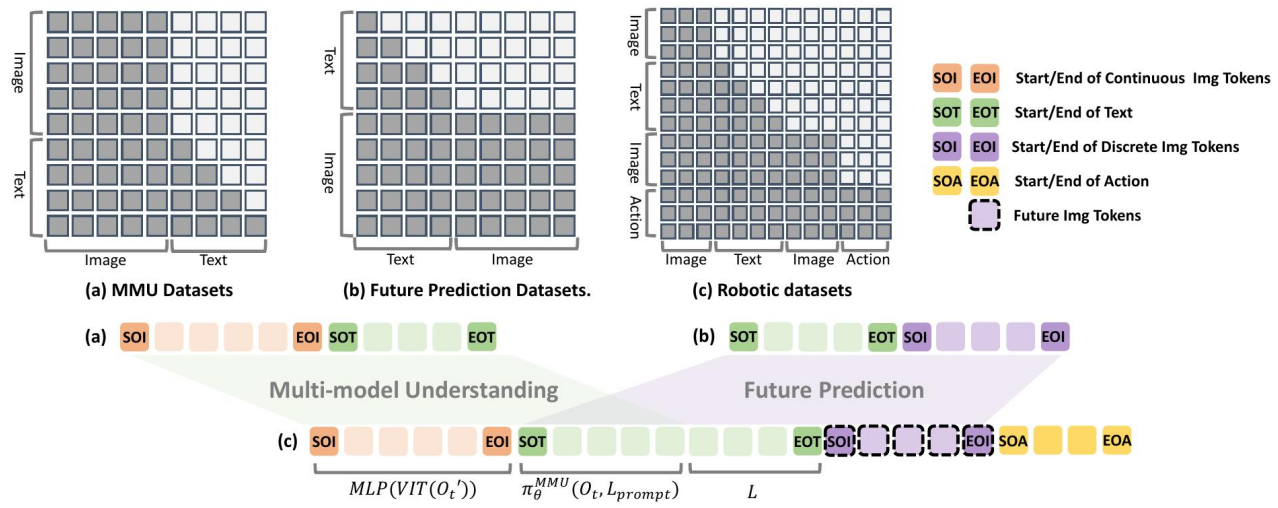
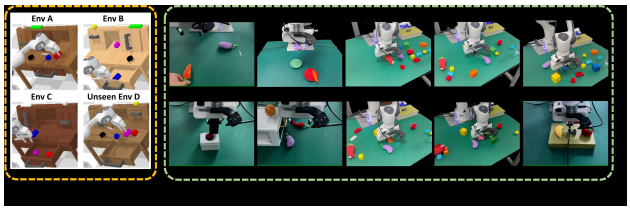
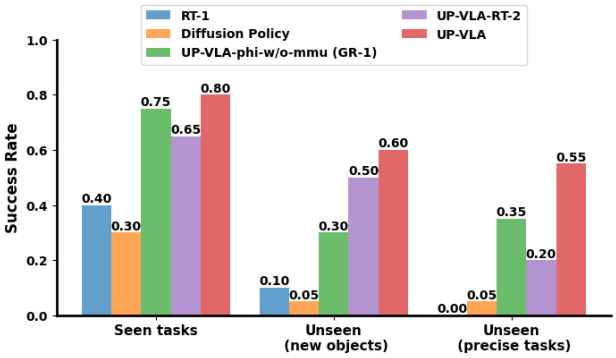


Figure 4. Unified prompting and attention in UP-VLA. A task token determines whether the model runs in multimodal understanding mode or future-prediction-and-action mode. The attention mask exposes image content differently for the two tasks while retaining a shared transformer backbone.



(a) Evaluation environments.



(b) Real-world manipulation results.

Figure 5. Real-world Panda evaluation. The full model achieves 0.80 success on seen tasks and 0.58 on unseen-object tasks, showing that joint understanding and prediction transfer beyond simulation.

Taken together, these trends suggest a clear division of labor inside the unified model. Predictive supervision mostly determines whether the policy preserves precise, object-centric state information across chained subtasks, which explains the large CALVIN drop when prediction is removed. MMU contributes more strongly when the robot must bind language to objects whose appearance differs from the robot data, so its benefit is most visible in the unseen-object real-world setting rather than the narrower CALVIN distribution.

One further pattern is that seen-task success is not monotonic with CALVIN performance. The model without MMU reaches 0.85 seen-task success, higher than the full model's 0.80, but it collapses on unseen objects. This suggests that the full objective is not merely improving an in-distribution control score; it is reshaping the representation toward robustness under appearance shift, which is the setting where semantic grounding is most needed.

5.2. Qualitative behavior

Figure 6 provides qualitative evidence for the intended behavior of the model. The MMU branch can produce scene descriptions and answer visual questions, while the predictive branch generates plausible future observations under the task instruction. This supports the interpretation that unseen-object performance benefits disproportionately from MMU training, while CALVIN gains are driven more strongly by prediction.

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABC→D	0.533	0.222	0.094	0.038	0.013	0.90
Diffusion Policy*	other	ABC→D	0.402	0.123	0.026	0.008	0.000	0.56
3D Diffuser Actor	other	ABC→D	**0.938**	0.803	0.662	0.533	0.412	3.35
3D-VLA	VLA	ABC→D	0.447	0.163	0.081	0.016	0.000	0.71
UP-VLA-RT-2*	VLA	ABC→D	0.612	0.389	0.236	0.138	0.062	1.44
Robo-Flamingo	VLA	ABC→D	0.824	0.619	0.466	0.331	0.235	2.47
Uni-Pi	Prediction	ABC→D	0.560	0.160	0.080	0.080	0.040	0.92
Susie	Prediction	ABC→D	0.870	0.690	0.490	0.380	0.260	2.69
GR-1	Prediction	ABC→D	0.854	0.712	0.596	0.497	0.401	3.06
UP-VLA-phi-w/o-mmu*	Prediction	ABC→D	0.844	0.705	0.604	0.520	0.430	3.13
UP-VLA	Prediction&VLA	ABC→D	0.928	**0.865**	**0.815**	**0.769**	**0.699**	**4.08**

Table 1. Zero-shot long-horizon evaluation on CALVIN with five chained tasks. Results marked with an asterisk (*) are our reproductions; the rest are copied from the original papers. UP-VLA achieves the best performance.

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABCD→D	0.844	0.617	0.438	0.323	0.227	2.45
Robo-Flamingo	VLA	ABCD→D	**0.964**	0.896	0.824	0.740	0.660	4.09
GR-1	Prediction	ABCD→D	0.949	0.896	0.844	0.789	0.731	4.21
UP-VLA	Prediction&VLA	ABCD→D	0.962	**0.921**	**0.879**	**0.842**	**0.812**	**4.42**

Table 2. Zero-shot long-horizon evaluation on the Calvin ABCD→D benchmark. Results of baselines are copied from original papers.

Method	ABC→D	Real World	
	Avg.Len↑	Seen↑	Unseen↑
w/o MMU	3.89	0.85	0.20
w/o Bridge-Pretrain	2.74	0.65	0.30
w/o Prediction	1.44	0.65	0.35
w/o MMU-Condition	3.99	0.80	0.50
Full	4.08	0.80	0.58

Table 3. Ablating components of UP-VLA.

6. Conclusion

UP-VLA shows that multimodal understanding and future prediction can be trained jointly inside a single autoregressive embodied policy. By combining a CLIP-based understanding stream, a discrete future-prediction stream, and an action head under unified prompting and attention, the model improves both long-horizon CALVIN performance and real-world unseen-object manipulation. The empirical pattern is consistent across main results and ablations: semantic grounding matters most for generalization, predictive supervision matters most for precise control, and the best embodied behavior emerges when both are optimized together.

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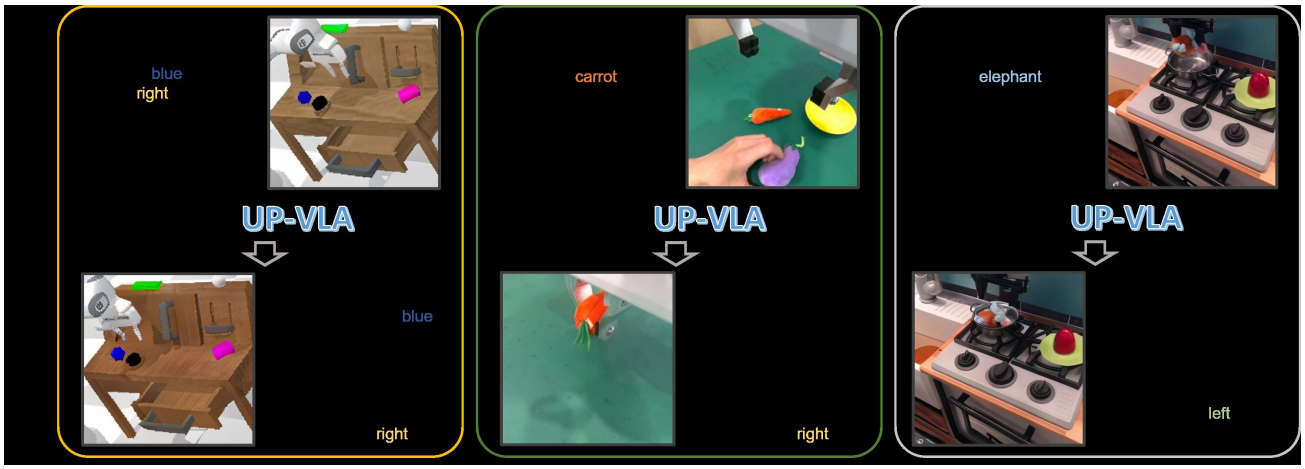

Figure 6. Qualitative examples of multimodal understanding and future prediction. The MMU branch answers scene questions, while the predictive branch generates future images aligned with the commanded manipulation.

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